FINM3013 Introduction to Financial Derivatives

Chapter 5 Exercises

Solutions

1.

The present value of the storage costs for nine months are

$$0.12 + 0.12e^{-0.10\times0.25} + 0.12e^{-0.10\times0.5} = 0.351$$

or \$0.351. The futures price is from equation (5.11) given by F_0 where

$$F_0 = (30 + 0.351)e^{0.1\times0.75} = 32.72$$

i.e., it is \$32.72 per ounce.

2. The theoretical forward exchange rate is $1.0404e^{(0.0025-0)\times0.25} = 1.041$.
If the actual forward exchange rate is 1.03, an arbitrageur can a) borrow X Swiss francs, b) convert the Swiss francs to 1.0404X dollars and invest the dollars for three months at 0.25% and c) buy X Swiss francs at 1.03 in the three-month forward market. In three months, the arbitrageur has $1.0404Xe^{0.0025\times0.25} = 1.041X$ dollars. A total of 1.3X dollars are used to buy the Swiss francs under the terms of the forward contract and a gain of 0.011X is made.
If the actual forward exchange rate is 1.05, an arbitrageur can a) borrow X dollars, b) convert the dollars to X/1.0404 Swiss francs and invest the Swiss francs for three months at zero interest rate, and c) enter into a forward contract to sell X/1.0404 Swiss francs in three months. In three months the arbitrageur has X/1.0404 Swiss francs. The forward contract converts these to (1.05X)/1.0404=1.0092X dollars. A total of $Xe^{0.0025\times0.25} = 1.0006X$ is needed to repay the dollar loan. A profit of 0.0086X dollars is therefore made.

3.

a) The present value, I, of the income from the security is given by:

$$I = 1\times e^{-0.08\times2/12} + 1\times e^{-0.08\times5/12} = 1.9540$$

The forward price, F_0, is given by:

$$F_0 = (50 - 1.9540)e^{0.08\times0.5} = 50.01$$

or \$50.01. The initial value of the forward contract is (by design) zero. The fact that the forward price is very close to the spot price should come as no surprise. When the compounding frequency is ignored the dividend yield on the stock equals the risk-free

rate of interest.

b) In three months:

$$I = e^{-0.08\times 2/12} = 0.9868$$

The delivery price, K, is 50.01. From equation (5.6) the value of the short forward contract, f, is given by

$$f = -(48 - 0.9868 - 50.01e^{-0.08\times 3/12}) = 2.01$$

and the forward price is

$$(48 - 0.9868)e^{0.08\times 3/12} = 47.96$$

4. Suppose that F_0 is the one-year forward price of the commodity. If F_0 is relatively high, the trader can borrow \$950 at 6%, buy one ounce of the commodity and enter into a forward contract to sell the commodity in one year for F_0. The profit made in one year is

$F_0 - 950\times 1.06 = F_0 - 1007$

This is profitable if F_0 >1007. If F_0 is relatively low, the trader can sell one ounce of the commodity for \$949, invest the proceeds at 5.5%, and enter into a forward contract to buy it back for F_0. The profit (relative to the position the trader would be in if the commodity were held in the portfolio during the year) is

$949\times 1.055 - F_0 = 1001.195 - F_0$

This shows that there is no arbitrage opportunity if the forward price is between \$1001.195 and \$1007 per ounce.